Case Study: Smart City Traffic Incident Detection System

1. System Context

Background

A city government deploys a **Smart Traffic Incident Detection System** to improve road safety and reduce congestion.
The system continuously collects data from:

- Roadside cameras
- Traffic sensors (speed, volume)
- GPS data from public buses and taxis
- Citizen reports (mobile app)

Its goal is to **detect traffic incidents in near real time** (accidents, stalled vehicles, congestion hotspots) and notify:

- Traffic control operators
- Emergency services
- Navigation systems

Functional Requirements

- Ingest high-volume, heterogeneous traffic data
- Clean and normalize incoming data
- Detect incidents (rule-based + AI) (with various algorithms/methods)
- Generate alerts with severity levels (with various algorithms/methods)
- Store data for historical analysis

Non-Functional Requirements

- Scalability (city-wide deployment)
- Low latency
- Extensibility (new detection algorithms)
- Reliability